

Introduction & Logistics

BGP Deployment Workshop

29th September – 2nd October 2025

Ulaanbaatar Hotel, Mongolia



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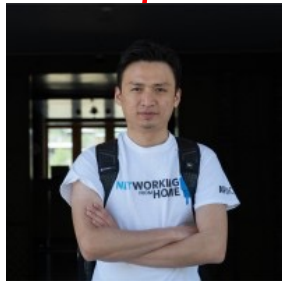
Last update 28th September 2025



Sponsors

- Network Startup Resource Center (NSRC)
- USA National Science Foundation (NSF)
- University of Oregon
- Mongolia Network Operators Group

Instructor Team



Name	Organisation (Country)
Philip Smith	NSRC (AU)
Tashi Phuntsho	FLEXOPTIX/NSRC (AU)
Karma Jamyang	DrukREN/GovTech (BT)
Bayanbaatar Batdorj	Mongolsat Networks (MN)
Batmagnai Erdene	Mobicom (MN)



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Logistics

- This is a 4-day workshop
 - Participation every day is required
 - Each day's content builds on the previous day
 - Completion certificates will be given to those who attend all four days
- All timings are Mongolia time (+8 UTC)
 - Please arrive on time!
- Please no social media!
- Mobile phones OFF or on SILENT
 - If you need to receive or make calls, please do so OUTSIDE the room

Logistics – WiFi

- Wireless Network
 - SSID for this workshop is NSRC-PFS
 - Network operates at both 2.4GHz and 5GHz
 - The key is “8888888888” – ten times the number 8

Logistics

- The workshop website:
 - <https://nsrc.org/workshops/2025/mnnog7/nsrc-mnnog7-bgp/>
 - This site has the agenda, all the presentation and lab materials, and any other relevant information for the workshop
- The Lab Environment
 - <https://vtp.ws.nsrc.org>
 - Where you will be connecting for access to the routers in the lab network
 - Keep this page available once the labs start



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The lab environment

Browser address bar: <http://www.ws.nsrc.org> | 120% | DNS | Search

NSRC Network Startup Resource Center **BGP Deployment**

Agenda | Downloads | NOC ▾ | Archived workshops

Group 1

Console ▾

- Router1
- Router2
- Router3

Group 2

Console ▾

Group 3

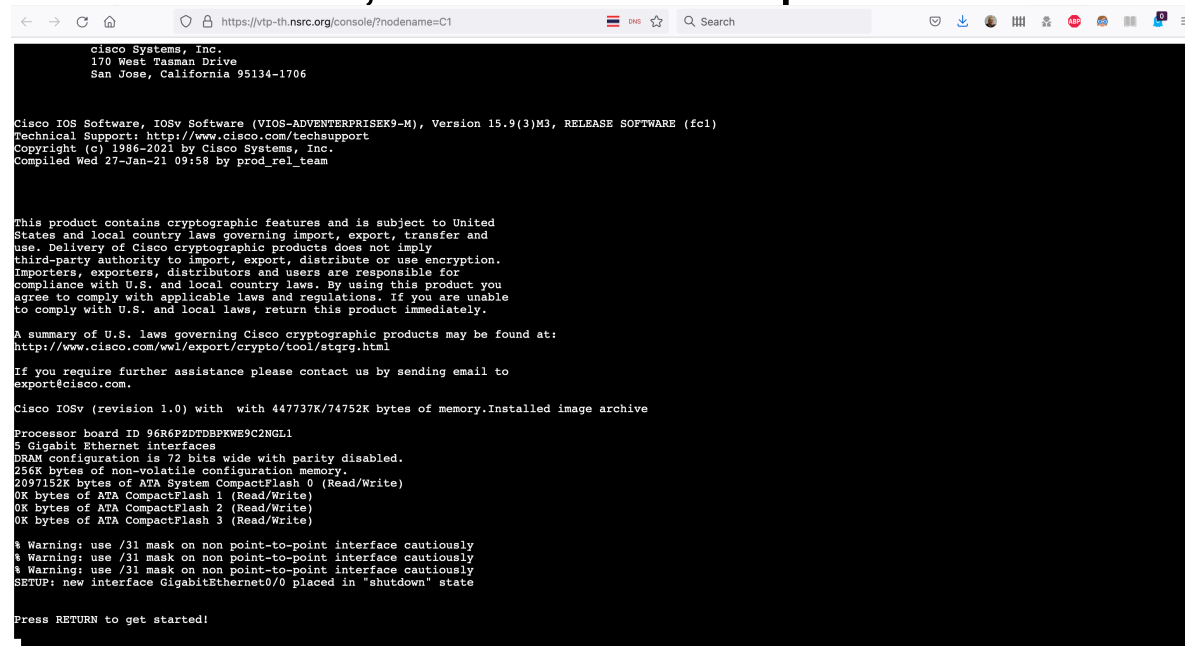
Console ▾

Group 4

Console ▾

The lab environment

- Once a console is selected, browser will open a new tab
 - For example:

A screenshot of a web browser window displaying a Cisco IOSv console interface. The browser's address bar shows the URL 'https://vtp-th.nsrc.org/console/?nodename=C1'. The console output is as follows:

```
cisco Systems, Inc.  
170 West Tasman Drive  
San Jose, California 95134-1706  
  
Cisco IOS Software, IOSv Software (VIOS-ADVENTERPRISEK9-M), Version 15.9(3)M3, RELEASE SOFTWARE (fcl)  
Technical Support: http://www.cisco.com/techsupport  
Copyright (c) 1986-2021 by Cisco Systems, Inc.  
Compiled Wed 27-Jan-21 09:58 by prod_rel_team  
  
This product contains cryptographic features and is subject to United  
States and local country laws governing import, export, transfer and  
use. Delivery of Cisco cryptographic products does not imply  
third-party authority to import, export, distribute or use encryption.  
Importers, exporters, distributors and users are responsible for  
compliance with U.S. and local country laws. By using this product you  
agree to comply with applicable laws and regulations. If you are unable  
to comply with U.S. and local laws, return this product immediately.  
  
A summary of U.S. laws governing Cisco cryptographic products may be found at:  
http://www.cisco.com/wvl/export/crypto/tool/stqrg.html  
  
If you require further assistance please contact us by sending email to  
export@cisco.com.  
  
Cisco IOSv (revision 1.0) with 447737K/74752K bytes of memory. Installed image archive  
Processor board ID 96R6P2DTDBPKWE9C2NGL1  
5 Gigabit Ethernet interfaces  
DRAM configuration is 72 bits wide with parity disabled.  
256K bytes of non-volatile configuration memory.  
2097152K bytes of ATA System CompactFlash 0 (Read/Write)  
0K bytes of ATA CompactFlash 1 (Read/Write)  
0K bytes of ATA CompactFlash 2 (Read/Write)  
0K bytes of ATA CompactFlash 3 (Read/Write)  
  
% Warning: use /31 mask on non point-to-point interface cautiously  
% Warning: use /31 mask on non point-to-point interface cautiously  
% Warning: use /31 mask on non point-to-point interface cautiously  
SETUP: new interface GigabitEthernet0/0 placed in "shutdown" state  
  
Press RETURN to get started!
```

- Simply press “return” key to continue and get the router command prompt



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Week Schedule

Day	Topic
Monday	Introduction to Routing, IS-IS, BGP, OSPF vs IS-IS
Tuesday	Scaling BGP, BGP Attributes, IBGP & EBGP
Wednesday	BGP Best Practices, Routing Filtering
Thursday	BGP Policy, Troubleshooting



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Day Schedule

Time	Activity
09:00-10:30	Morning Session 1
10:30-11:00	Break
11:00-12:30	Morning Session 2
12:30-13:30	Lunch
13:30-15:30	Afternoon Session 1
15:30-16:00	Break
16:00-17:30	Afternoon Session 2



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Routing Configuration Practice

- The environment is a Virtual Environment
 - Virtual routers (GNS3) & PCs (Linux Containers)
- You will use your web browser to access these devices from your laptop
 - Make sure you have the latest version of Chrome or Edge or Firefox or Safari installed on your laptop
- You will be working in groups



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Key for Success

- Remember, this workshop is for you!
 - If you do not understand or have questions, ask for clarification
 - Please share your thoughts and experiences. They are valuable
 - Take advantage of the event to meet your colleagues and build the “human network”



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Questions?